Finiteness of Shallow Logical Operators and a Sharp Depth Threshold

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For a quantum code on n finite-dimensional physical subsystems with distance d, we consider all exact logical unitaries realized by circuits of depth at most h and circuit gates acting on at most ρ subsystems. For a uniform upper-bound $h < 2\lceil \log_\rho d \rceil - 1$, these circuits can realize only finitely many logical actions up to a global phase, without any stabilizer or geometric locality assumptions. In contrast, for every qubit stabilizer code encoding at least one qubit, a non-trivial continuous one-parameter family of logical unitaries is available at depth $2\lceil \log_\rho d \rceil - 1$.

Controlling the error propagation is the central question in fault-tolerant computation. A single fault in a depth-h circuit with gate range ρ can affect at most ρ^h subsystems. So shallow quantum circuits of bounded range can effectively limit error propagation between error-correction steps. Under the constant-depth and bounded-range premise, Gottesman's threshold construction protects polynomial-size computations below a noise threshold with constant space overhead [1]. Nevertheless, the constant-depth premise seems not sufficient for complete, or even infinitely many logical operations. The celebrated Eastin–Knill theorem [2] shows that transversal gates with respect to a fixed partition into detectable blocks realize only finitely many logical actions up to a global phase. The same restriction holds for constant-depth circuits with geometric locality or algebraic assumptions [3–6]. Therefore, one may ask at what depth this necessary condition can first hold.

As we will show in the following, the precise threshold depth is $2\lceil \log_\rho d \rceil - 1$, given $d, \rho \geq 2$. More specifically, Theorem 1 rules out an infinite logical image when $h < 2\lceil \log_\rho d \rceil - 1$, without any stabilizer or geometric locality assumption. And Theorem 2 shows that every qubit stabilizer code encoding at least one qubit realizes a nontrivial continuous logical family at depth $2\lceil \log_\rho d \rceil - 1$, when arbitrary gates on at most ρ qubits are allowed. Stabilizer subspace codes are thus the most capable candidates for the emergence of an infinite logical family, attaining the smallest possible depth at given d and ρ.

Settings and notations. Let the physical system be $\mathcal{H} = \bigotimes_{i=1}^n \mathcal{H}_i$, with $n \geq 1$ and finite-dimensional factors $\mathcal{H}_i$. The code $\mathcal{C}$ is specified by an isometry $V : \mathcal{H}_L \otimes \mathcal{H}_G \longrightarrow \mathcal{H}$, where $\dim \mathcal{H}_L = K \geq 2$ and $\dim \mathcal{H}_G = g \geq 1$ denote the dimensions of logical and gauge subspaces, respectively. A distance lower bound d means that every operator O supported on fewer than d physical subsystems satisfies

$$V^\dagger O V = I_L \otimes M_O \tag{1}$$

for some operator M_O on $\mathcal{H}_G$ [7, 8]. A physical unitary U implements an exact logical action when

$$UV = V(W \otimes Z) \tag{2}$$

for unitaries W on $\mathcal{H}_L$ and Z on $\mathcal{H}_G$. This equality holds for all logical and gauge states, and the logical action is the phase class $[W] \in \mathrm{PU}(K)$. For integers $h \geq 1$ and $\rho \geq 2$, let $\mathcal{L}_{\leq h}(\mathcal{C}, \rho)$ be the set of these classes obtained when U has at most h layers of arbitrary unitary gates on disjoint supports of at most ρ physical subsystems. The depth bound applies to each complete circuit.

To prove the finiteness of $|\mathcal{L}_{\leq h}(\mathcal{C}, \rho)|$, we generalize the argument in the Eastin–Knill Theorem [2] and covariant error correction [9, 10] to circuit families that need not form groups. For each arrangement, the polynomial circuit map [11] and encoding condition define a real algebraic set. Finite smooth cell decomposition [12, Cor. 8.4], proved here in Lemma 1, reduces finiteness to constancy of the projective logical action on each cell. Along a smooth path within a cell, we differentiate the circuit into gate insertions. The depth bound makes each insertion's forward or backward support detectable. Error detection makes each compression gauge-only in a common frame, using the encoded action for the backward terms. The logical generator is therefore scalar, so each cell contributes at most one logical action up to phase.

Theorem 1. *Let $n, h \geq 1$, and $\rho \geq 2$ be integers, and let V encode a quantum code $\mathcal{C}$ satisfying Eq. (1) for a distance $d \geq 2$ on n physical subsystems. If*

$$h < 2\lceil \log_\rho d \rceil - 1, \tag{3}$$

then the logical actions of all circuits of depth at most h, with gates on at most ρ subsystems, that satisfy Eq. (2) obey

$$|\mathcal{L}_{\leq h}(\mathcal{C}, \rho)| < \infty.$$

All gate supports and unitary matrices are allowed. Consequently, $\mathcal{L}_{\leq h}(\mathcal{C}, \rho)$ is not dense in $\mathrm{PU}(K)$.

Proof. Equation (3) is equivalent to $\rho^{\lceil \frac{h}{2} \rceil} < d$. After we add singleton identity gates on idle sites and pad to h layers, each layer is a partition of the n sites into blocks of

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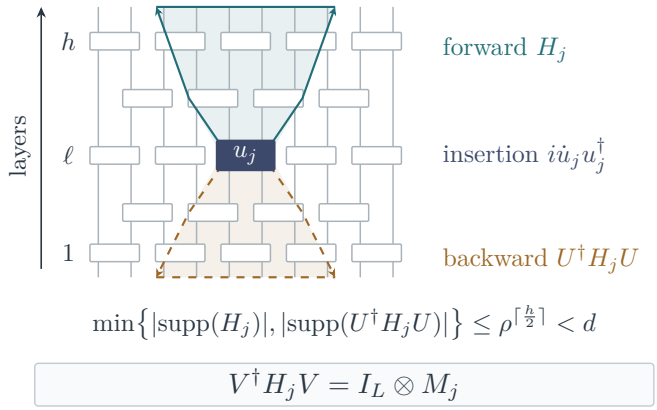


FIG. 1. Two-sided light cones. The variation of gate u_j in layer ℓ gives the propagated generator H_j. Under the depth condition, its forward support or the backward support of $U^\dagger H_j U$ is detectable. Error detection and the exact encoded action yield gauge-only compression of H_j in the output frame. The brickwall arrangement assumes no geometric locality.

size at most ρ. There are finitely many such partitions, so it suffices to prove finiteness for one support arrangement α.

Fix α and list its gates in layer order as $u_1, \dots, u_J$, with $U = u_J \cdots u_1$. Let D_j be the Hilbert-space dimension on the support of gate j. All admissible gate matrices and encoded factors are represented in

$$\mathcal{P}_\alpha = \left(\prod_{j=1}^{J} \mathrm{U}(D_j) \right) \times \mathrm{U}(K) \times \mathrm{U}(g).$$

Let $\mathcal{X}_\alpha \subseteq \mathcal{P}_\alpha$ consist of the tuples $(\mathbf{u}, W, Z)$ satisfying Eq. (2). Unitarity and the encoding condition are polynomial equations in the real and imaginary matrix entries, so $\mathcal{X}_\alpha$ is a real algebraic set.

The finite cell decomposition theorem [12, Cor. 8.4], in the smooth form proved in Lemma 1 in the End Matter, gives

$$\mathcal{X}_\alpha = \bigsqcup_{r=1}^{N_\alpha} C_r, \qquad N_\alpha < \infty, \tag{4}$$

where each cell is smoothly diffeomorphic to an open cube $(0,1)^{\nu_r}$, with a point allowed when $\nu_r = 0$. The partition includes singular points of $\mathcal{X}_\alpha$. It remains to show that $[W]$ is constant on every C_r.

Take two tuples $x, y \in C_r$, choose a smooth diffeomorphism $\varphi_r : (0,1)^{\nu_r} \to C_r$, and set $a = \varphi_r^{-1}(x)$, $b = \varphi_r^{-1}(y)$. They are joined within C_r by

$$\gamma(t) = \varphi_r\big((1-t)a + tb\big), \qquad 0 \le t \le 1. \tag{5}$$

The coordinates of γ supply smooth gates $u_j(t)$ and factors $W(t), Z(t)$ satisfying Eq. (2) throughout. To prove that $[W(t)]$ is independent of t, we will show that the Hermitian generator of the physical path has gauge-only compression. With dots denoting derivatives in t, the product rule gives

$$H = i\dot{U}U^\dagger = \sum_{j=1}^{J} U_{>j}\big(i\dot{u}_j u_j^\dagger\big)U_{>j}^\dagger = \sum_{j=1}^{J} H_j, \tag{6}$$

where $U_{>j} = u_J \cdots u_{j+1}$, and $H_j = U_{>j}(i\dot{u}_j u_j^\dagger)U_{>j}^\dagger$ is the variation of gate j propagated through the later gates. If gate j is in layer ℓ, its insertion acts on at most ρ sites. Each later layer can enlarge its support by at most a factor ρ, giving a forward support of at most $\rho^{h-\ell+1}$ sites. To obtain a detectable term even when this bound is large, consider the same variation in the input frame,

$$U^\dagger H_j U = U_{<j}^\dagger (i u_j^\dagger \dot{u}_j) U_{<j},$$

where $U_{<j} = u_{j-1} \cdots u_1$. Its backward support has at most ρ^ℓ sites. Other gates in layer ℓ have disjoint supports and cancel from both counts. Thus at least one of the two supports has size at most

$$\rho^{\min\{\ell, h-\ell+1\}} \le \rho^{\lceil \frac{h}{2} \rceil} < d.$$

We now compress every summand in the same output frame. If the forward support is smaller than d, Eq. (1) applies directly to H_j. Otherwise the backward support is smaller than d, so $V^\dagger U^\dagger H_j U V = I_L \otimes N_j$. The exact encoded action in Eq. (2) transfers this to

$$\begin{aligned} V^\dagger H_j V &= (W \otimes Z)(I_L \otimes N_j)(W^\dagger \otimes Z^\dagger) \\ &= I_L \otimes Z N_j Z^\dagger. \end{aligned}$$

In either case $V^\dagger H_j V = I_L \otimes M_j$. Summing gives the desired gauge-only compression

$$V^\dagger H V = I_L \otimes M \tag{7}$$

with $M = \sum_{j=1}^{J} M_j$ Hermitian on $\mathcal{H}_G$.

To translate this restriction into constancy of $[W(t)]$, differentiate Eq. (2) and use $U^\dagger V = V(W^\dagger \otimes Z^\dagger)$. This gives

$$\begin{aligned} V^\dagger H V &= A \otimes I_G + I_L \otimes B, \\ A = i\dot{W}W^\dagger, &\qquad B = i\dot{Z}Z^\dagger. \end{aligned} \tag{8}$$

Comparing Eqs. (7) and (8) and taking the partial trace over $\mathcal{H}_G$ yields

$$A = \frac{\mathrm{Tr}(M - B)}{g} I_L.$$

Thus $A = \lambda(t) I_L$ for real $\lambda(t)$, and $\dot{W} = -i\lambda(t)W$. Integrating gives $W(1) = \exp\big(-i \int_0^1 \lambda(t)\, dt\big) W(0)$, so the two arbitrary tuples $x, y \in C_r$ have the same projective logical action.

Each cell therefore contributes at most one logical action up to phase, and α contributes at most N_α such actions. Taking the union over the finitely many support arrangements proves that $\mathcal{L}_{\le h}(\mathcal{C}, \rho)$ is finite. Since $K \ge 2$, this finite set is closed and cannot be dense in $\mathrm{PU}(K)$. □

Finiteness does not exclude a finite subset $S \subseteq \mathcal{L}_{\le h}(\mathcal{C}, \rho)$ for which

$$\overline{\langle S \rangle} = \mathrm{PU}(K).$$

The group $\langle S \rangle$ allows words of arbitrary finite length. A word of length r in $S \cup S^{-1}$ has a concatenated implementation of depth at most rh. These products are not required to remain in $\mathcal{L}_{\le h}(\mathcal{C}, \rho)$, so density of the generated group does not assert density within the original depth bound. For example, Dua et al. give a $[[37, 1, 7]]$ CSS code with separate depth-one circuits of gate range $\rho = 3$ that implement [13, Sec. V and Appendix B 3]

$$S = \{\overline{T}^\dagger, \sqrt{\overline{X}}^\dagger\} \subseteq \mathcal{L}_{\le 1}(\mathcal{C}, 3).$$

The two circuits use different partitions of the physical qubits. Since $\overline{H} \propto (\overline{T}^\dagger)^2 \sqrt{\overline{X}}^\dagger (\overline{T}^\dagger)^2$ and $\overline{T} = (\overline{T}^\dagger)^7$, this pair generates a dense subgroup of PU(2). For this code, $2\lceil \log_3 7 \rceil - 1 = 3$, so $\mathcal{L}_{\le 1}(\mathcal{C}, 3)$ is finite by Theorem 1.

In a fault-tolerance threshold construction based on the constant-depth unitary logical gadgets considered here, logical-gate compilation is necessary for universal control in the finite-image regime. On a fixed code, the available shallow logical actions form a finite alphabet. If its generated group is dense, arbitrary target logical unitaries can be approximated by words over this alphabet, with error correction interleaved as prescribed by the protocol [1, 13]. The constant-depth premise applies to each gadget. Compiled word lengths cannot remain uniformly bounded when arbitrary targets must be approximated to arbitrarily high accuracy.

The finiteness condition has an exact boundary for stabilizer codes. For an integer $\rho \ge 2$, define the critical depth

$$h_\infty(\mathcal{C}, \rho) = \min\{h \in \mathbb{Z}_{\ge 1} : \mathcal{L}_{\le h}(\mathcal{C}, \rho) \text{ is infinite}\}, \quad (9)$$

with value ∞ if the set is empty. When $d \ge 2$, Theorem 1 gives

$$h_\infty(\mathcal{C}, \rho) \ge 2\lceil \log_\rho d \rceil - 1. \quad (10)$$

For the upper bound of $h_\infty(\mathcal{C}, \rho)$, we implement continuous rotations generated by a weight-d logical Pauli. A Clifford tree collects its nonidentity factors onto at most ρ qubits in $\lceil \log_\rho d \rceil - 1$ layers. The compressed Pauli then fits in one allowed rotation gate. Conjugating back recovers the original Pauli rotation, at total depth $2\lceil \log_\rho d \rceil - 1$. This uses the established tree constructions for parity compression, phase gadgets, and Pauli-rotation synthesis [14–16]. The resulting continuous family is nonconstant modulo phase, giving the matching upper bound.

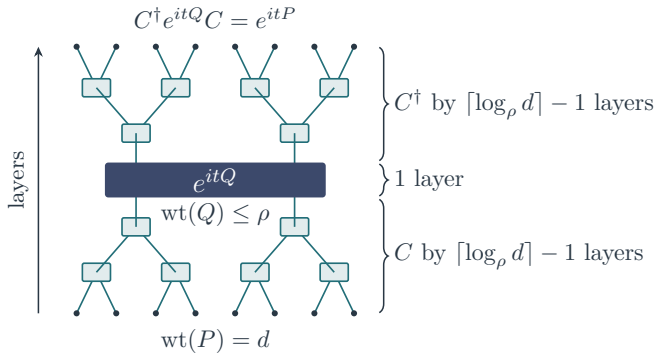


FIG. 2. Mechanism of Theorem 2. Time runs upward. C compresses the support of a weight-d logical Pauli to $Q = CPC^\dagger$, with $\mathrm{wt}(Q) \le \rho$. One rotation gate followed by $C^\dagger$ gives the complete logical circuit $C^\dagger e^{itQ} C = e^{itP}$, of depth $2\lceil \log_\rho d \rceil - 1$.

Theorem 2. *Let $\mathcal{C}$ be an $[[n, k, d]]$ qubit stabilizer code, with $k \ge 1$ and actual distance $d \ge 2$. Let $\rho \ge 2$ be an integer, and allow arbitrary unitary gates on any set of at most ρ qubits. Then*

$$h_\infty(\mathcal{C}, \rho) = 2\lceil \log_\rho d \rceil - 1. \quad (11)$$

At this depth, the logical image contains a nontrivial continuous one-parameter subgroup.

Proof. Put $m = \lceil \log_\rho d \rceil$. It suffices to construct infinitely many logical actions at depth at most $2m - 1$. Choose a Hermitian code-preserving Pauli P of weight d, with $P^2 = I$ and nonscalar logical action $\overline{P}$ [17, Sec. 3.2]. We first compress its support so that its rotation fits in one allowed gate.

We adapt the unitary parity-concentration part of the binary-tree in-line symmetry-verification circuit in Ref. [14, Sec. IV and Fig. 2(a)] to gates on at most ρ qubits. At each layer, partition the active factors into groups of ρ, with at most one smaller group. In each group, basis changes and CNOTs compose into one allowed Clifford gate collecting the product onto one qubit. The disjoint groups form one layer. If w_r is the weight after r layers, then $w_0 = d$ and $w_{r+1} = \lceil \frac{w_r}{\rho} \rceil$, giving $w_r = \lceil \frac{d}{\rho^r} \rceil$. Thus $m - 1$ layers yield C with $Q = CPC^\dagger$ of weight at most ρ. The compressed rotation e^{itQ} is one allowed gate. Conjugating back gives

$$U(t) = C^\dagger e^{itQ} C = e^{itP}. \quad (12)$$

The depth is at most $(m - 1) + 1 + (m - 1) = 2m - 1$. The complete circuit preserves the code and induces $e^{it\overline{P}}$. As $\overline{P}$ has both eigenvalues ± 1, its projective rotations form a nontrivial continuous U(1) subgroup of period π. Hence $h_\infty(\mathcal{C}, \rho) \le 2m - 1$, and the lower bound gives Eq. (11). □

The construction attains the threshold for an infinite exact logical image, but does not by itself provide fault tolerance against arbitrary single-gate faults. An error $e^{i\varepsilon Q}$ immediately after the central rotation gate propagates through $C^\dagger$ to $e^{i\varepsilon P}$, which induces a nontrivial logical error for $\varepsilon \notin \pi\mathbb{Z}$.

Conclusion. For general quantum codes in the finite-dimensional setting considered here, bounded-range circuits below the stated threshold realize only finitely many exact logical actions up to phase. The proof combines a detectable light cone in one of the two circuit directions with a finite smooth decomposition of the admissible circuit parameters. Qubit stabilizer codes attain the bound by concentrating the support of a logical Pauli into one rotation gate and then reversing the compression. Together, the theorems locate the critical circuit depth to support infinitely many exact logical actions without imposing geometric locality.

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Lean proofs of the statements of Theorems 1 and 2, under their stated hypotheses, together with verification scripts, are available in the project repository (proof snapshot `411ff86`). The formal proof of Theorem 1 uses an independent algebraic argument; it does not formalize the CAD construction in the End Matter.

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END MATTER

Finite smooth decomposition of polynomial zero sets

Together with the infinitesimal argument of Theorem 1, finite Whitney stratification of the semialgebraic constraint set into connected smooth strata already gives a finite logical image. To keep the argument elementary and self-contained, we give a short proof using linear algebra and the implicit function theorem, with elementary polynomial arithmetic. The finite decomposition below is the polynomial case of standard semialgebraic theory [12]. A smooth cell below means an embedded smooth submanifold that is smoothly diffeomorphic to an open cube $(0,1)^r$, with a point allowed when $r = 0$. No smoothness of the full zero set is assumed.

Lemma 1. *For any finite family of real polynomials on $\mathbb{R}^N$, there is a finite partition of $\mathbb{R}^N$ into smooth cells on each of which every polynomial has constant sign. Consequently, the common zero set of finitely many real polynomials is a finite disjoint union of smooth cells.*

Proof. We induct on N. In one variable, the finitely many real roots of the nonzero polynomials partition the line into points and open intervals. The assertion is also immediate for $N = 0$. Write the variables in the induction step as (x, t), where $x \in \mathbb{R}^{N-1}$. Discard identically zero input polynomials, and write the remaining ones as $p_1(x,t), \dots, p_a(x,t)$.

We first specify a finite family of projection polynomials in x. For every nonempty subset $I \subseteq \{1, \dots, a\}$, form $F_I = \prod_{i\in I} p_i$ and include all its coefficients in t. For every truncation

$$f(x,t) = \sum_{j=0}^{m} f_j(x) t^j$$

of F_I, with $1 \le m \le \deg_t F_I$, also include all square minors of the coefficient matrix of the linear map

$$T_f : (A, B) \longmapsto Af + B\partial_t f, \qquad \deg A \le m-1, \quad \deg B \le m. \tag{13}$$

The target consists of polynomials of degree at most $2m-1$, so this matrix has $2m$ rows and $2m+1$ columns. Its entries and minors are polynomials in x. Zero polynomials may be discarded. There are finitely many subsets, truncations, coefficients, and minors. By induction, partition the base $\mathbb{R}^{N-1}$ into finitely many smooth cells on which these projection polynomials have constant signs.

Fix one base cell C. Each p_i is either identically zero as a polynomial in t at every $x \in C$, or has one constant degree there. This follows from the coefficient signs. If all are identically zero, the cylinder $C \times \mathbb{R}$ is already a suitable cell. Otherwise let I be the indices of the nonzero factors and set $F = F_I$. Its degree m is constant on C. If $m = 0$, the same cylinder suffices. Suppose $m \geq 1$. The truncation at this degree agrees with F on every fiber over C, so its projected matrix represents T_F.

The projected minors make the rank of T_F constant on C. At a point x, write $F = d_0 f_0$, $\partial_t F = d_0 g_0$, where d_0 is the monic greatest common divisor and f_0, g_0 are coprime. Put $s = \deg d_0$ and $r = m - s$. Then $\deg f_0 = r$, $\deg g_0 = r-1$, and $r \geq 1$. The identity $AF + B\partial_t F = 0$ holds exactly when $A = Qg_0$ and $B = -Qf_0$. The degree restrictions in Eq. (13) give $\deg Q \leq s$. Hence

$$\dim \ker T_F = s+1, \qquad \operatorname{rank} T_F = 2m - s.$$

In particular, s and r are constant on C.

The monic squarefree part $G = \frac{f_0}{c}$, where c is the leading coefficient of F, has precisely the real roots of F, each simple: differentiation lowers each root multiplicity by one. We show that its coefficients depend smoothly on $x \in C$, even where F has repeated roots. Solve

$$\begin{aligned} AF + B\partial_t F = 0, \qquad & \deg A < r, \\ B = t^r + \sum_{j<r} b_j t^j. \qquad & \end{aligned} \tag{14}$$

There is exactly one solution: $A = -\frac{g_0}{c}$, $B = \frac{f_0}{c}$. The coefficient matrix for the $2r$ unknown coefficients is a column submatrix of T_F. It has full column rank: a homogeneous solution would have $\deg B < r$, whereas any nonzero kernel vector has $B = -Qf_0$ of degree at least r. Choose a nonzero $2r$-by-$2r$ minor of this coefficient matrix at one point of C. It is among the projected minors and thus remains nonzero throughout C. Cramer's rule expresses every coefficient in Eq. (14) as a rational function of x with a denominator that does not vanish on C. Thus $G = B$, the monic squarefree part, has smooth coefficients on C. The same construction for the singleton products gives a smooth monic squarefree part G_i for every nonconstant active factor p_i.

The number of real roots of G is locally constant on C. Indeed, at a point x_0, its real roots are simple. Apply the implicit function theorem in a smooth coordinate chart of C: each root persists uniquely and smoothly. A monic polynomial $t^r + \sum_{j<r} b_j t^j$ has all its real roots in $|t| \leq 1 + \max_j |b_j|$. Near x_0, this gives one uniform bound. Outside small disjoint intervals about the old roots, $G(x_0, t)$ is bounded away from zero on the remaining compact interval. Continuity excludes new roots there. Within the small intervals, the derivative stays nonzero and there is exactly one root. Since C is connected, the number of real roots is constant, and ordering them gives globally smooth functions

$$\theta_1(x) < \cdots < \theta_b(x).$$

Inactive factors vanish identically, and active constant factors have constant nonzero signs. For an active nonconstant factor, suppose $p_i(x_0, \theta_j(x_0)) = 0$. The simple root of G_i at that point persists as a smooth root $\beta(x)$. Because p_i divides F, this is also a root of G, and the local uniqueness of the latter root gives $\beta(x) = \theta_j(x)$. Thus vanishing on the graph is an open condition in C. Nonvanishing is open by continuity. Connectedness makes one alternative hold everywhere, and any nonzero sign is then constant.

Partition $C \times \mathbb{R}$ into these b root graphs and the $b+1$ intervening open bands, including the unbounded bands. No active factor vanishes in a band, so its sign is constant there. A graph is diffeomorphic to C. A band is diffeomorphic to $C \times (0,1)$. For finite endpoints use affine interpolation. For a half-line use $\frac{u}{1-u}$, and for the whole line use $\frac{u}{1-u} - \frac{1-u}{u}$, where $0 < u < 1$. Thus all pieces are smooth cells. There are finitely many base cells, and each yields finitely many pieces. This completes the induction. The common zero set is the union of those cells on which all the given polynomials vanish. □

To apply the lemma, express the unitary and encoding constraints for one circuit-support arrangement in real and imaginary matrix entries. Their common zero set is $\mathcal{X}_\alpha$. For two points in one cell, a smooth chart from an open cube carries the straight segment between their preimages to a smooth path in $\mathcal{X}_\alpha$. The circuit calculation in Theorem 1 gives $\dot{W} = -i\lambda W$ along this path, so its endpoints have the same projective logical action. Each cell contributes at most one such action. Finitely many cells, followed by finitely many support arrangements, give the claimed finiteness.